Mira Askar

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EDUCATION

University of Warwick — **Coventry, United Kingdom**
BEng Electrical and Electronic Engineering — Sep 2025 - Present
Expected graduation: July 2028 | First-year average: 74.8% (First Class) | Engineering Mathematics: 91%

Malvern College — Sep 2023 - Aug 2025
A Levels: Mathematics (A), Physics (A), Russian (A*)

TECHNICAL SKILLS

Languages: Java, SQL, C/C++, MATLAB
Frameworks & Testing: Spring Boot, Hibernate ORM, JUnit
Tools & Technologies: PostgreSQL, Docker, Git, REST APIs, JSON, Simulink
Hardware & Embedded: ESP32, STM32, Arduino, KiCad, LTspice, CAN, UART

WORK EXPERIENCE

Healware — **Hong Kong (Remote)**
Java Backend Developer — Aug 2026 - Sep 2026

- Built a prototype Java/Spring Boot REST API (4 business endpoints + 1 technical endpoint) that ingests medication selections and patient allergy histories as JSON and returns allergy-risk assessments.
- Implemented the risk logic: direct active-ingredient matching plus rule-based cross-reactivity checks, returning contributing factors and safer medication alternatives through structured DTOs.
- Delivered the service independently with input validation, exception handling and structured error responses (correct HTTP status codes), tested with JUnit and integration HTTP tests on synthetic patient data.

Trans Asia Construction LLP — **Kyzylorda, Kazakhstan**
Java Backend Developer — Jul 2026 - Aug 2026

- Developed a Java/Spring Boot prototype for employee record management using Hibernate ORM and PostgreSQL, containerised with Docker.
- Designed a relational database schema and REST APIs for employee CRUD operations, structured search and document-status tracking, supporting a proposed transition from Excel-based record keeping.
- Implemented server-side validation, database constraints and transactional updates to maintain consistency across employee records.

PROJECTS

Poker Decision Simulator — Mar 2026
Java, JUnit, Monte Carlo Simulation

- Built a Java Texas Hold'em simulator with complete betting rounds, hand evaluation and three rule-based opponents with distinct playing styles.
- Implemented Monte Carlo hand-equity estimation using 5,000 simulations per decision to support decisions under incomplete information.
- Developed an explainable decision engine using pot odds and estimated call EV, with rule-based raise recommendations and JUnit tests covering hand rankings, betting rules and pot distribution.

Self-Balancing Robot — Jun 2026
University Engineering Project | MATLAB, Simulink, Arduino Mega 2560, PID Control, LaTeX

- Tuned the PID balance controller on live hardware via Simulink Monitor & Tune, using binary search over gains: cut derivative gain (0.5 → 0.15) to stop overcorrection, raised integral gain to reject disturbances and added ±255 motor saturation.
- Sustained balance for 15+ minutes on carpet, a wooden desk and the lab floor, 30× the 30-second target.
- Co-authored the project documentation in LaTeX (Overleaf), tracing the V-Model from requirements to validation; project awarded First Class.

Warwick Hyperloop — Oct 2025 - Present
Electronics Hardware Engineer, Battery Subteam | KiCad, LTspice, STM32, CAN, UART

- Designed a battery-temperature monitoring PCB in KiCad (schematic and layout) around an STM32 microcontroller, reading 8 temperature sensors through multiplexers.
- Simulated the design in LTspice before prototyping, validating voltage dividers, multiplexer switching and diode/transistor stages.
- Integrated CAN and UART interfaces into the PCB design to connect the board with the BMS and the Control Systems subteam's Arduino Mega.

ACHIEVEMENTS & INTERESTS

Awards: British Physics Olympiad (BPhO) - Gold; Gold CREST Award (fuel-level sensor system for QinetiQ).
Scholarships: University of Warwick Global Excellence Scholarship; Malvern College Honorary Scholarship.
Sponsorship: Secured approximately £1,000 from KT Lab Cloud and Chevron at the Enactus National Expo Kazakhstan.
Interests: Piano (completed music school with a final grade 5.0/5.0); choir and orchestra; Warwick volleyball; robotics.